Deliverable 1 - DragonEndPOS

Matthew Arsenault

101288386

Team 84

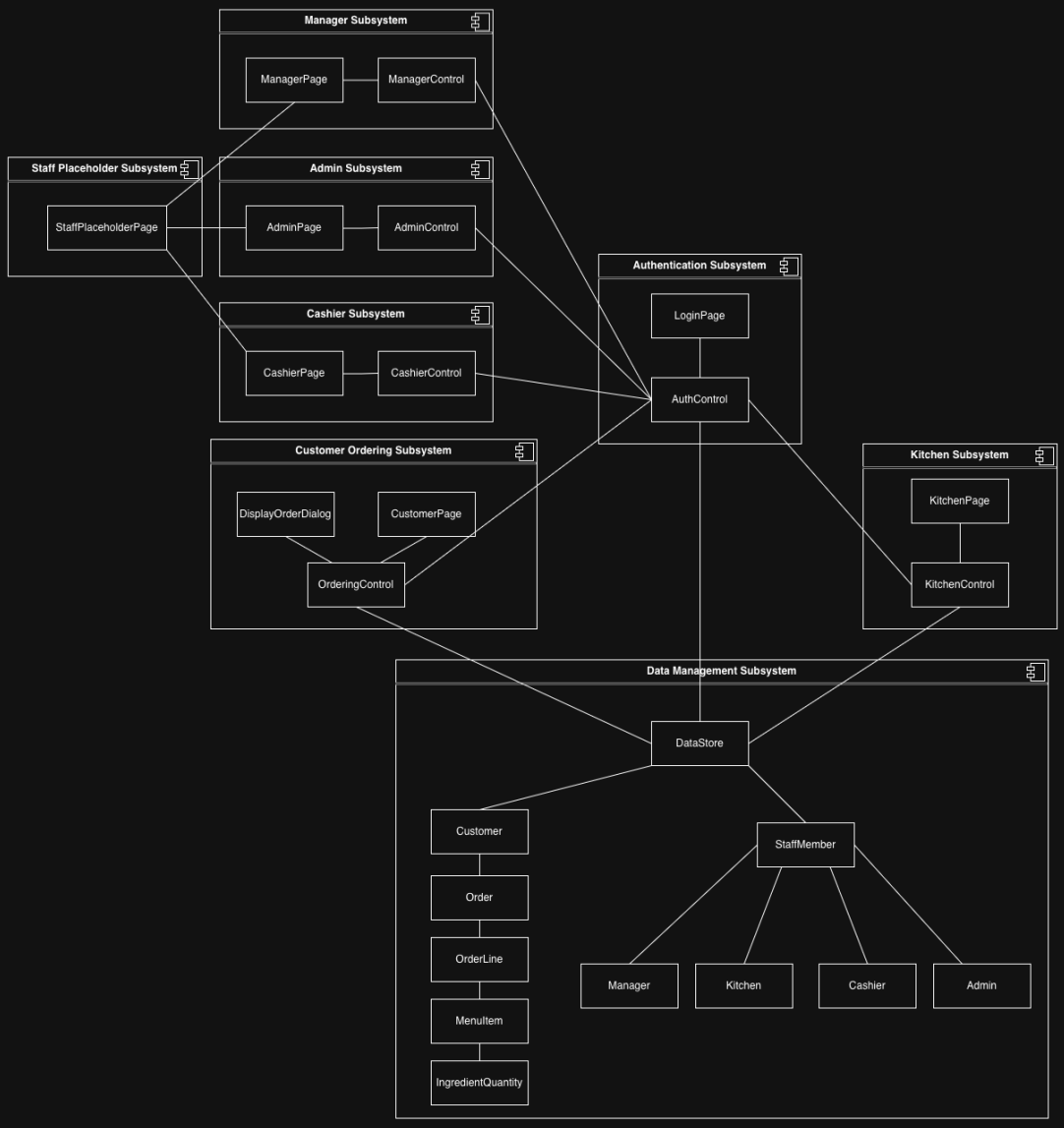
Submitted to:

Dr. Christine Laurendeau

COMP 3004 - Object-Oriented Software Engineering

School of Computer Science

Carleton University


Manager Subsystem
ManagerPage
ManagerControl
Staff Placeholder Subsystem
StaffPlaceholderPage
Admin Subsystem
AdminPage
AdminControl
Authentication Subsystem
LoginPage
AuthControl
Cashier Subsystem
CashierPage
CashierControl
Customer Ordering Subsystem
DisplayOrderDialog
CustomerPage
OrderingControl
Kitchen Subsystem
KitchenPage
KitchenControl
Data Management Subsystem
DataStore
Customer
Order
OrderLine
MenuItem
IngredientQuantity
StaffMember
Manager
Kitchen
Cashier
Admin

The system uses a repository based system where feature based subsystems, each used by a type of user, can access data from the Data Management subsystem named DataStore. This is done because each feature shares data but none of them need to talk to each other, apart from a dependency on login authentication, to complete their function. This was chosen over an N-Tier system as intermediary tiers are yet to be implemented and clients share a single data store.

Coupling is minimized by separating the responsibilities of each user into their respective subsystem and allowing them to access a single point to retrieve data. A change in a subsystem will not affect the other subsystems since they only interact through DataStore and don't reference each others classes. Clients also only use the entities their feature needs and all access goes through the repository. If a client wanted to add a new staff role, this would only interact with the data management and authentication classes and not the other features.

The cohesion in each subsystem is high as each controls aspects that only their features needs. The data storage stores the systems shared data between two groups, orders and staff, with no links between them.

The boundary classes exist as the login page, customer page, kitchen page, display order page, and placeholder pages for roles that have yet to be implemented. Each boundary class has a control class associated with it, where the control logic currently exists within the boundary class itself.